



2025 IMAGEOMICS DIGITAL DIGEST NEWSLETTER ISSUE 5



Celebrating 75 Years of NSF

Seventy-five years ago, the U.S. National Science Foundation (NSF) was created to promote scientific progress, national health, prosperity, and security. Since 1950, it has fueled transformative research across disciplines, serving as a cornerstone of American innovation.

NSF's investment in basic science has led to breakthroughs—from the early internet to quantum computing—helping position the U.S. as a global leader in research and technology. Its commitment to merit-based funding and interdisciplinary collaboration continues to drive solutions to society's greatest challenges.

One key effort, the NSF Harnessing the Data Revolution (HDR) initiative, supports cutting-edge methods for analyzing vast datasets. By advancing data science tools and infrastructure, HDR equips researchers to extract insights from complex information and solve real-world problems.

The Imageomics Institute, supported by HDR, embodies this mission by applying machine learning to biological images to reveal traits and patterns that deepen our understanding of biodiversity and evolution. This data-driven approach bridges biology and AI, supporting both discovery and conservation. As we mark NSF's 75th anniversary, we highlight this enduring impact. With NSF's support, institutions like Imageomics are pushing the frontiers of science—and shaping a more informed, connected, and sustainable future.

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AUGUST 18-21, 2025

**A COLLABORATIVE HANDS-ON MACHINE
LEARNING EVENT FOR EXTRACTING
FUNCTIONAL TRAITS FROM IMAGES.**



Decker to Defend PhD with a Focus on Bats and Big Data

Long after the sun sets, Sydney Decker's research continues—unraveling the evolutionary story of bats, whose wings carry secrets written in genomes and traits.

On June 18, she defends her Ph.D. at The Ohio State University, marking a major milestone in a career driven by curiosity, quantitative thinking, and a love for the world's most misunderstood creatures: bats. Decker, a researcher in the

Department of Evolution, Ecology and Organismal Biology (EEOB),

was first introduced to bat biology during her undergraduate studies, earning a research scholarship, and getting paired with a faculty mentor specializing in bat evolution. That partnership sparked what would become nearly a decade of research into the evolutionary innovations and ecological diversity of bats.

“After learning about the evolutionary innovations and ecological diversity in bats, I decided to continue studying them throughout my undergraduate degree and for my PhD research,” Decker said. “Evolution is arguably the most complex process that can be studied, so I wanted to ensure I was making the most of the data I could collect and using sound data practices throughout my research.”



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ACCOLADES

YU SU PROMOTED TO ASSOCIATE PROFESSOR AT OHIO STATE



The Ohio State University Board of Trustees promoted Dr. Yu Su to Associate Professor in Computer Science and Engineering, citing his excellence in teaching, research, and service. Su leads the Machine Learning Foundations team at the NSF-funded Imageomics Institute, where he develops AI tools to extract biological meaning from images. His research focuses on large language models and language agents with reasoning and planning capabilities. A recipient of the Sloan Research Fellowship and multiple university honors, Su's promotion highlights his leadership in AI and his impact on interdisciplinary science at Ohio State.

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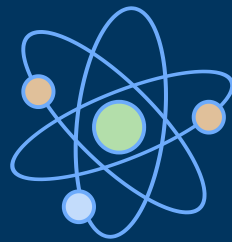


Haifa Keynote: Tanya Berger-Wolf



Imageomics Institute Director Tanya Berger-Wolf was the keynote speaker at the recent University of Haifa AI Conference, held May 25-27 in Isreal. She spoke on the topic of #AIforNature.

Haifa University President Gur Alroey speaks at the opening ceremony.



Chain Reaction:



Spotlight: Otto Brookes



Chain Reaction is a spotlight story series where one rising researcher lights the path to the next. Follow the chain and meet the minds shaping the future of Imageomics.

When a colleague described Otto Brookes' work as "pure Imageomics," he highlighted a research endeavor epitomizing the fusion of biology and machine learning. Brookes, a PhD candidate at the University of Bristol, is at the forefront of developing AI systems that extract meaningful behavioral traits from images and videos of wild apes, directly supporting the Imageomics Institute mission.

Brookes nominated the next Chain Reaction story profile to focus on Jenna Kline from The Ohio State University.

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Karpatne Named Faculty Fellow



Imageomics colleague Anuj Karpatne was named Faculty Fellow in the Virginia Tech College of Engineering. This award is given in "recognition of extraordinary performance in research." Pictured here with Anuj during the Dean's Awards Celebration, are Dean Julia Ross (left) and Christine Julien, head of the Virginia Tech Department of Computer Science.



NextGen Maruf Earns Ph.D. Award

Imageomics NextGen, M. Maruf, received the 2025 Outstanding Ph.D. Research Award from the Virginia Tech Department of Computer Science. He played a pioneering role in KGML and Multi-modal Vision-Language Models for scientific discovery as part of the Imageomics Institute. He is now an Applied Scientist at Amazon.



Generative AI on Track to Shape the Future of Drug Design

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Using advanced artificial intelligence, researchers have developed a novel method to make drug development faster and more efficient.

In a new paper, [Xia Ning](#), lead author of the study and a professor of [biomedical informatics](#) and [computer science and engineering](#) at The Ohio State University, introduces DiffSMol, a generative AI model capable of generating realistic 3D structures of small molecules that can serve as promising drug candidates. DiffSMol works by analyzing the shapes of known ligands – molecules that bind to protein targets – and using these shapes as conditions to generate novel 3D molecules that better bind to the protein targets. Study results showed that when used to create molecules with the potential to quicken the drug-making process, DiffSMol has a 61.4% success rate, outperforming prior research attempts that achieved success about 12% of the time.

“By using well-known shapes as a condition, we can train our model to generate novel molecules with similar shapes that don’t exist in previous chemical databases,” said Ning.

The study was published in [Nature Machine Intelligence](#).

Story from Ohio State News



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Have Imageomics news to share? Tell us through the [newsletter submission form](#), or contact Ryan Horns in Communications at: Horns.1@osu.edu